

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Seventh Semester B.E. (Mechanical) (Credit System) Degree Examinations

January 2023

19MEE31 - WIND AND SOLAR POWER ENGINEERING

Duration: 3 Hours

Max. Marks: 100

Note: 1) Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

2) Use of solar table is permitted.

Unit – I

Unit – I		Marks	BT*	CO*	PO*
1.	a) Define the following terms: Latitude, Altitude, Declination.	6	L*1	1	1
	b) With a neat sketch explain the working of sunshine recorder.	4	L2	1	1
	c) With a neat sketch explain central receiver power plant.	10	L2		1
2.	a) With a neat schematic diagram explain the working of a solar pond.	10	L2	1	2
	b) Estimate the monthly average daily global radiation on a horizontal surface at Vadodara ($22^{\circ} 00'N$, $73^{\circ} 10'E$) on March 16, if the average sunshine hours per day is 9.5. Assume $a = 0.28$ and $b = 0.48$ for the location.	10	L3	1	1
3.	a) With a neat sketch explain the solar industrial process heating system using a concentrating collector array.	10	L2	2	1
	b) Calculate the angle made by beam radiation with the normal to a flat-plate collector on May 1 at 0900hrs (LAT). The collector is located in New Delhi ($28^{\circ} 35'N$, $77^{\circ} 12'E$) and is tilted at an angle of 36° with the horizontal and is pointing due south.	10	L2	1	1

Unit – II

4.	a)	Explain with relevant diagram the principle of working of a solar cell.	10	L2	3	1
	b)	With a neat sketch explain the working of wind assisted gas turbine generating unit.	10	L2	4	1
5.	a)	Explain current-voltage and power-voltage characteristics of a solar cell.	10	L2	3	1
	b)	What are the factors that determine the output from a wind energy converter? Derive an expression for the power available in the wind.	10	L2	4	1
6.	a)	Explain the parameters to be considered while selecting site for wind turbine.	10	L2	4	1
	b)	With a neat sketch explain a photovoltaic water-pumping system.	10	L2	3	1

Unit – III

7.	a)	With a schematic diagram explain various components of a horizontal axis wind turbine.	10	L2	5	1
	b)	Explain aerodynamic forces acting on the blade with the help of a vector diagram of forces.	10	L2	5	1
8.	a)	Discuss the environmental effects of wind turbine.	10	L2	5	1
	b)	Wind at 1 atmospheric pressure and 15°C has a velocity of 15 m/s. Calculate:				
	i)	The total power density in the wind stream.				
	ii)	The maximum obtainable power density.				
	iii)	A reasonably obtainable power density.				
	iv)	The total power and				
	v)	The torque.				
		Consider propeller type wind turbine.				
		Take gas turbine diameter = 120 m and turbine operating speed = 40 rpm at maximum efficiency.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
